

Mohamed Khalil ZRELLY

Generative AI Engineer Apprentice | Computer Science Engineering Student

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Professional Summary

Generative AI Engineer Apprentice at Crédit Agricole, specializing in agentic systems and RAG pipelines, and Engineering student at Sup Galilée, Sorbonne Paris North University, with solid foundations in algorithms and problem-solving.

Work Experience

Crédit Agricole Personal Finance & Mobility

France

Generative AI Engineer Apprentice

September 2025 — Present · 10 months

- Developed a declarative, config-driven microservice for **workflow management and orchestration**, making it easy to build, manage, and execute complex AI workflows at scale.
- Built an automated **workflow evaluation engine** leveraging **LLM-as-a-judge** to assess outputs against custom metrics, with a **dashboard** to analyse workflow behaviour across different scenarios.

Agentic Workflows · Advanced RAG · Amazon Bedrock · LLM Observability · Python/FastAPI · Amazon DynamoDB · Streamlit

VERMEG for Banking & Insurance Software

Tunisia

Natural Language Processing Intern

February 2024 — June 2024 · 5 months

- Implemented a system that automates email classification and information extraction from attachments, using **BERT** and **OCR** to optimise efficiency and processing time.
- Developed a web application for **BERT Fine-Tuning**, email management, and real-time data analysis and visualisation, to enhance user interaction and performance monitoring.

Hugging Face Transformers · Sentence-Transformers · Fine-Tuning · PyTorch · Tesseract OCR · Python/Flask · MongoDB · React/Redux

Sofiatech

Tunisia

Machine Learning Intern

June 2023 — August 2023 · 3 months

- Evaluated multiple **deep reinforcement learning** algorithms from **Stable Baselines 3** across various **Gymnasium** environments to assess agent performance and training efficiency.
- Built a **2D grid game** in **Pygame** where a **Deep Q-Learning** agent learns to navigate a dynamic environment, applying the best-performing algorithm from the benchmarking phase.

Reinforcement Learning · Deep Q-Learning · stable-baselines3 · TensorFlow/Keras · Gymnasium · NumPy · Pandas · Matplotlib · Python/Pygame

Featured Projects

CVision | RAG · CV Recommendation System

June 2025 — August 2025

Created an intelligent **CV recommendation system** based on job description using **Retrieval-Augmented Generation (RAG)**, to optimise the selection and automatic ranking of the most relevant profiles.

Advanced RAG · Re-Ranking · LangChain · Qwen3-VL · Qdrant · Python/Streamlit

Twixel | Multi-Agent · Social Listening System

In Development — 2026

Designing an intelligent **multi-agent system** for automated social listening on X (Twitter), aimed at identifying emerging trends in real time and generating actionable marketing campaign ideas at scale.

Multi-Agent Systems · LangGraph · MCP/A2A · GPT-OSS · Qdrant · Python/FastAPI · Vue.js

Technical Skills

Machine Learning, DL & MLOps	PyTorch, TensorFlow, Scikit-Learn, XGBoost, Amazon SageMaker
Generative AI & Agentic Systems	LangChain, LangGraph, Milvus, Qdrant, Amazon Bedrock
Data Analysis & Engineering	Pandas, Matplotlib, PySpark, Apache Airflow, Amazon S3
SQL & NoSQL Databases	Oracle, MySQL, PostgreSQL, MongoDB, Amazon DynamoDB
Full-Stack Web Development	FastAPI, Flask, Express.js, Laravel, React, Vue.js
Version Control, Testing & QA	Git, GitHub, Pytest, Jest, Vitest, Playwright, Postman
Cloud Infrastructure & DevOps	AWS, Docker, Kubernetes, Terraform, GitHub Actions
Programming Languages	Python, JavaScript/TS, PHP, Java, C/C++, PL/SQL

Education

Sup Galilée – Sorbonne Paris North University

Villetaneuse, France

Engineering Degree in Computer Science

2024 — 2027 (Expected)

ISI Ariana – University of Tunis El Manar

Ariana, Tunisia

Bachelor of Science in Computer Science

2021 — 2024 · High Distinction

Certifications & Languages

Microsoft Azure	Azure AI Fundamentals (AI-900) — Jan 2024 · Azure Data Fundamentals (DP-900) — Jan 2024
Udemy	Spark and Python for Big Data with PySpark — Mar 2024 · NLP with Python — Feb 2024
Spoken Languages	Arabic (Native) · French B2 — TCF · English B2 — TOEIC